A. Orac and LCM

time limit per test: 3 seconds
memory limit per test: 256 megabytes

For the multiset of positive integers $s = \{s_1, s_2, \dots, s_k\}$, define the Greatest Common Divisor (GCD) and Least Common Multiple (LCM) of s as follow:

- $\gcd(s)$ is the maximum positive integer x, such that all integers in s are divisible on x.
- $\text{lcm}(s)$ is the minimum positive integer x, that divisible on all integers from s.

For example, $\gcd(\{8, 12\}) = 4, \gcd(\{12, 18, 6\}) = 6$ and $\text{lcm}(\{4, 6\}) = 12$. Note that for any positive integer x, $\gcd(\{x\}) = \text{lcm}(\{x\}) = x$.

Orac has a sequence a with length n. He come up with the multiset $t = \{\text{lcm}(\{a_i, a_j\}) \mid i < j\}$, and asked you to find the value of $\gcd(t)$ for him. In other words, you need to calculate the GCD of LCMs of all pairs of elements in the given sequence.

Input

The first line contains one integer n $(2 \le n \le 100\,000)$.

The second line contains n integers, $a_1, a_2, \dots, a_n$ $(1 \le a_i \le 200\,000)$.

Output

Print one integer: $\gcd(\{\text{lcm}(\{a_i, a_j\}) \mid i < j\})$.

Examples

input Copy
```
2
1 1
```
output Copy
```
1
```

input Copy
```
4
10 24 40 80
```
output Copy
```
40
```

input Copy
```
10
540 648 810 648 720 540 594 864 972 648
```
output Copy
```
54
```

Note

For the first example, $t = \{\text{lcm}(\{1, 1\})\} = \{1\}$, so $\gcd(t) = 1$.

For the second example, $t = \{120, 40, 80, 120, 240, 80\}$, and it's not hard to see that $\gcd(t) = 40$.

Codeforces Round 641 (Div. 1)
Finished

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→ Problem tags

data structures | math | number theory | *1600

No tag edit access

→ Contest materials

- Codeforces Round #641 (en)
- Tutorial (en)